4.5 Model Comparison: RMSE and Final Selection

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Goal

This section will compare forecasting performance across all methods using holdout RMSE and select the final model.

Methods included:

- Simple Rules (baseline models)
- Exponential Smoothing (ETS)
- ARIMA models
- ARMAX models

The primary evaluation metric is holdout RMSE, which reflects out-of-sample predictive accuracy.

RMSE Caluclations

The following computations complete RMSE calculations for models derived from previous analysis.

Note for the ARIMA approach, we will first apply RMSE calculations to all candidate ARIMA models to decide on a best-performing model with smallest RMSE.

Combining RMSE Results Across Explored Methods

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
y_df <- readr::read_csv(
  "../../data/processed/daily_store_a_full.csv",
  show_col_types = FALSE
) |>
  arrange(Date)

y_all <- y_df |> pull(Sales)
n_train <- floor(0.8 * length(y_all))

y_train <- y_all[1:n_train]
y_test  <- y_all[(n_train+1):length(y_all)]
holdout_idx <- (n_train + 1):length(y_all)

simple_rmse <- readr::read_csv(
  "../../04_analysis/simple/results/simple_rules_aggregate_holdout_rmse_summary.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(method = "Simple")

ets_rmse <- readr::read_csv(
  "../../04_analysis/exponential/results/ets_holdout_rmse_summary.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(method = "ETS")

agg <- readr::read_csv(
  "../../data/processed/store_type_a_train_aggregate_daily.csv"
) |>
  dplyr::mutate(Date = as.Date(Date)) |>
  dplyr::arrange(Date)
```

```
## Rows: 942 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl  (10): Sales, OpenStores, PromoStores, PromoShare, SchoolHolidayShare, D...
## date  (1): Date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
train_subset <- agg |> slice(1:n_train)
model1 <- auto.arima(ts(train_subset$Sales, frequency = 7))
model2 <- Arima(ts(train_subset$Sales, frequency = 7), order=c(4,0,1), seasonal=c(2,1,1))
model3 <- Arima(ts(train_subset$Sales, frequency = 7), order=c(4,0,1), seasonal=c(1,1,0))
model4 <- Arima(ts(train_subset$Sales, frequency = 7), order=c(2,0,1), seasonal=c(2,1,0))
model5 <- Arima(ts(train_subset$Sales, frequency = 7), order=c(4,0,2), seasonal=c(2,1,0))

models <- list(
  ARIMA_auto = model1,
  ARIMA_m2 = model2,
  ARIMA_m3 = model3,
  ARIMA_m4 = model4,
  ARIMA_m5 = model5
```

```r
)

mod <- auto.arima(y_train)
arima_rmse_table <- purrr::imap_dfr(models, function(mod, name) {

  fc <- forecast::forecast(mod, h = length(y_test))$mean

  rmse_val <- sqrt(mean(
    (y_test - fc)^2,
    na.rm = TRUE
  ))

  tibble(
    model = name,
    holdout_rmse = rmse_val,
    n_holdout = length(y_test),
    method = "ARIMA"
  )
})

arima_rmse <- arima_rmse_table |>
  dplyr::arrange(holdout_rmse) |>
  dplyr::slice(1)

# ARMAX (TEMP placeholder - will be added)
# armax_rmse <- readr::read_csv("...") |> mutate(method = "ARMAX")

rmse_all <- dplyr::bind_rows(
  simple_rmse,
  ets_rmse,
  arima_rmse
#  armax_rmse
)

rmse_all <- rmse_all |>
  dplyr::mutate(
    model_full = paste(method, model, sep = " - ")
  ) |>
  dplyr::arrange(holdout_rmse)

rmse_all
```

```
## # A tibble: 6 x 5
##   model                 holdout_rmse n_holdout method model_full
##   <chr>                        <dbl>     <dbl> <chr>  <chr>
## 1 iid_by_period             1013538.       189 Simple Simple - iid_by_period
## 2 ARIMA_m2                  1016497.       189 ARIMA  ARIMA - ARIMA_m2
## 3 ETS                       1422687.        42 ETS    ETS - ETS
## 4 persistence_by_period     1665219.       189 Simple Simple - persistence_by_p~
## 5 global_average            1784744.       189 Simple Simple - global_average
## 6 persistence               2516853.       189 Simple Simple - persistence
```

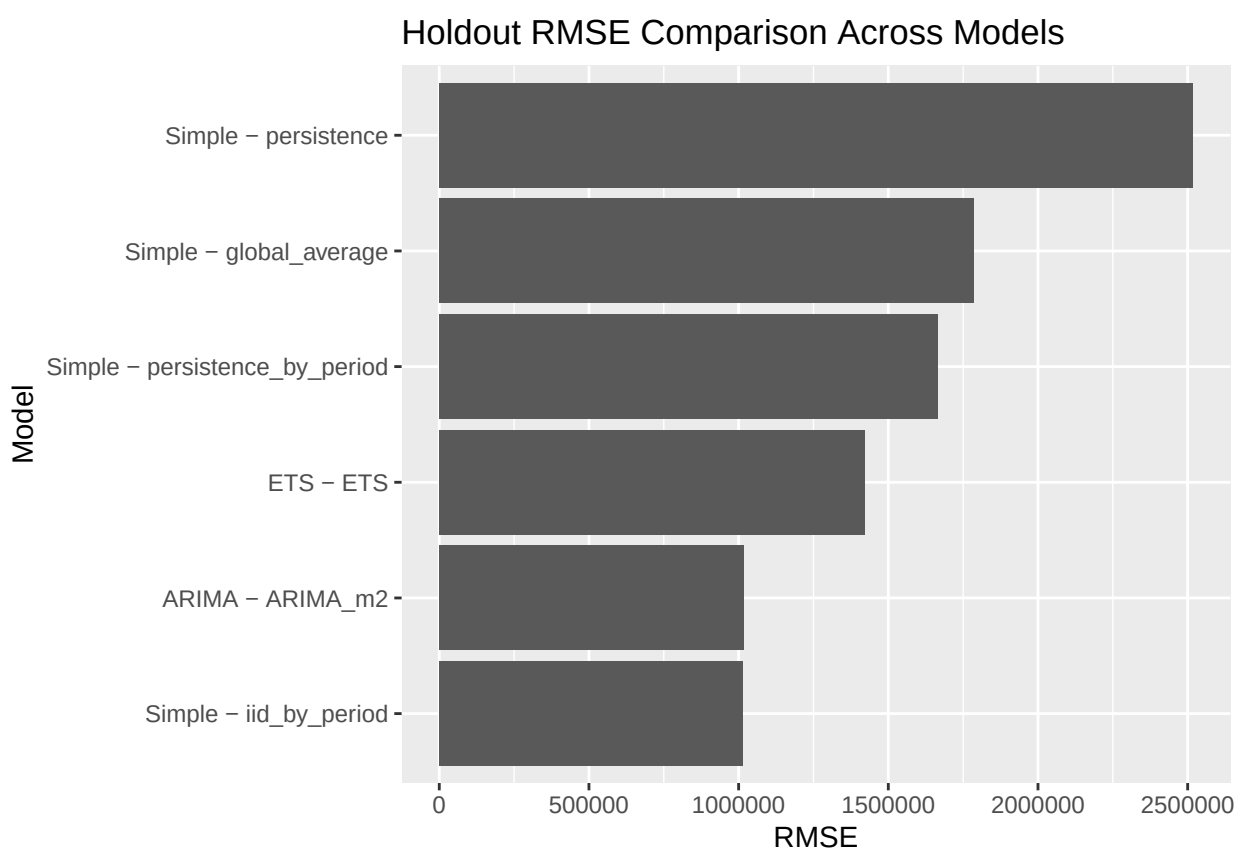
ARIMA Model Finalization: Based on the results above, the best ARIMA model out of all candidates is model 2 with RMSE = 1016497 (sale value). We will proceed with this model as a representative for the ARIMA method in the remaining analysis.

Interpretation Overall the table above compares forecasting performance across all finalized models, where:

- Lower RMSE indicates better predictive accuracy.
- Baseline models such as persistence provide a benchmark for evaluating more complex approaches.
- Improvements over simple rules indicate that the more complex models successfully capture additional structure in the data.

Let's visualize these comparisons.

```
ggplot(rmse_all, aes(x = reorder(model_full, holdout_rmse), y = holdout_rmse)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Holdout RMSE Comparison Across Models",
    x = "Model",
    y = "RMSE"
  )
```



Interpretation

The bar chart above highlights differences in forecasting performance:

- Models with substantially lower RMSE (e.g. ARIMA model 2) provide insights to how the target variable may be predicted.

- If complex models (ARIMA/ARMAX) only marginally outperform simple rules, their added complexity may not be justified. Here ARIMA significantly improves forecast accuracy compare to most simple baseline models, with the exception of Simple IID by Period method outperforming most approaches.
- Nonetheless, large improvements of more complex models overall suggest that time series structure and/or external variables are important in this scenario.

Final Model Selection

```
best_model <- rmse_all |>
  dplyr::slice_min(order_by = holdout_rmse, n = 1)

best_model
```

```
## # A tibble: 1 x 5
##   model         holdout_rmse n_holdout method model_full
##   <chr>                <dbl>     <dbl> <chr>  <chr>
## 1 iid_by_period     1013538.       189 Simple Simple - iid_by_period
```

Interpretation The selected model is the one with the lowest holdout RMSE, meaning the model demonstrates the best out-of-sample forecasting performance among the candidates.

Acknowledging Limitation: Evaluation Consistency

Currently, models were evaluated using different holdout strategies:

- Simple Rules, ARIMA and ARMAX: 80/20 split

- ETS: fixed 42-day holdout

This means RMSE values are not perfectly comparable across methods.

For a fully fair comparison, all models should be evaluated using:

- The same training period;

- The same holdout period;

- The same forecasting horizon.

This is recognized as a limitation in this project, and we wish to have the inconsistency addressed in future refinement.

Save Results

```r
dir.create("results", showWarnings = FALSE, recursive = TRUE)

readr::write_csv(rmse_all, "results/model_comparison_rmse.csv")
```